

PL: Assignment I

August 19, 2025

Q1. Let v be the valuation such that $v(p_1) = T$, $v(p_2) = F$, $v(p_3) = F$. Find $v(p_1 \rightarrow (\neg p_2 \wedge p_3))$.

Q2. Let A be the wff $((p_1 \rightarrow p_2) \wedge \neg p_1) \rightarrow \neg p_2$. Find a valuation that satisfies A and one that does not.

Q3. These puzzles are similar to some that appear in Smullyan's book *Forever Undecided*. The inhabitants of the planet GP are either green or purple. Green people always tell the truth and purple people never tell the truth. Consider two people named X and Y .

1. X says: *I am purple and Y is purple.* What color is X ? What color is Y ?
2. X says: *I am purple or Y is green.* What color is X ? What color is Y ?
3. X says: *If I am green, then Y is green.* What color is X ? What color is Y ?
4. X says: *I am green if and only if Y is green.* What color is Y ?

Q4.

1. Show that $p \rightarrow q$ and $\neg p \vee q$ have the same truth tables.
2. Find the truth table of $(\neg p \rightarrow q) \vee r$.

Q5. Complete the unfinished proofs of Propositions/Theorems in the lectures.

Q6.

1. Is $p \rightarrow (\neg p \rightarrow p)$ a tautology? What about $\neg p \rightarrow (\neg p \rightarrow p)$?
2. Show that if α , $\alpha \rightarrow \beta$ are tautologies, so is β .
3. Let α be a wff whose only connective is $\vee$. Can α be tautology? Justify your answer.

Q7. (Characterization of $\rightarrow$)

Show that each of the following is a tautology.

- $(\alpha \rightarrow \beta) \leftrightarrow (\neg\alpha \vee \beta)$.
- $(\alpha \rightarrow \beta) \leftrightarrow \neg(\alpha \wedge \neg\beta)$.
- $(\alpha \rightarrow \beta) \leftrightarrow (\neg\alpha \vee (\alpha \wedge \beta))$.

Q8. Construct a wff α which is neither a tautology nor a contradiction, and which makes the following wff true. p, q are propositional variables.

$$((\alpha \wedge q) \rightarrow \neg p) \rightarrow ((p \rightarrow \neg q) \rightarrow \alpha).$$

Q9. Show the following, where $\top$ denotes any tautology, and $\perp$ any contradiction.

1. $p \rightarrow q \equiv \neg p \vee q$
2. $p \equiv \neg\neg p$
3. $p \equiv p \wedge (p \vee q); p \equiv p \vee (p \wedge q)$
4. $p \rightarrow q \equiv \neg q \rightarrow \neg p$
5. $p \wedge \top \equiv p; p \vee \perp \equiv p$
6. $p \vee \top \equiv \top; p \wedge \perp \equiv \perp$
7. $p \vee p \equiv p; p \wedge p \equiv p$
8. $(p \vee q) \vee r \equiv p \vee (q \vee r); (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
9. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r); p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
10. $\neg(p \wedge q) \equiv \neg p \vee \neg q; \neg(p \vee q) \equiv \neg p \wedge \neg q$.

Q10. (a) Can two distinct propositional variables be logically equivalent?

(b) Prove that there is one equivalence class of wffs each with respect to the relation $\equiv$, containing

- (i) *exactly* the tautologies, and
- (ii) *exactly* the contradictions.